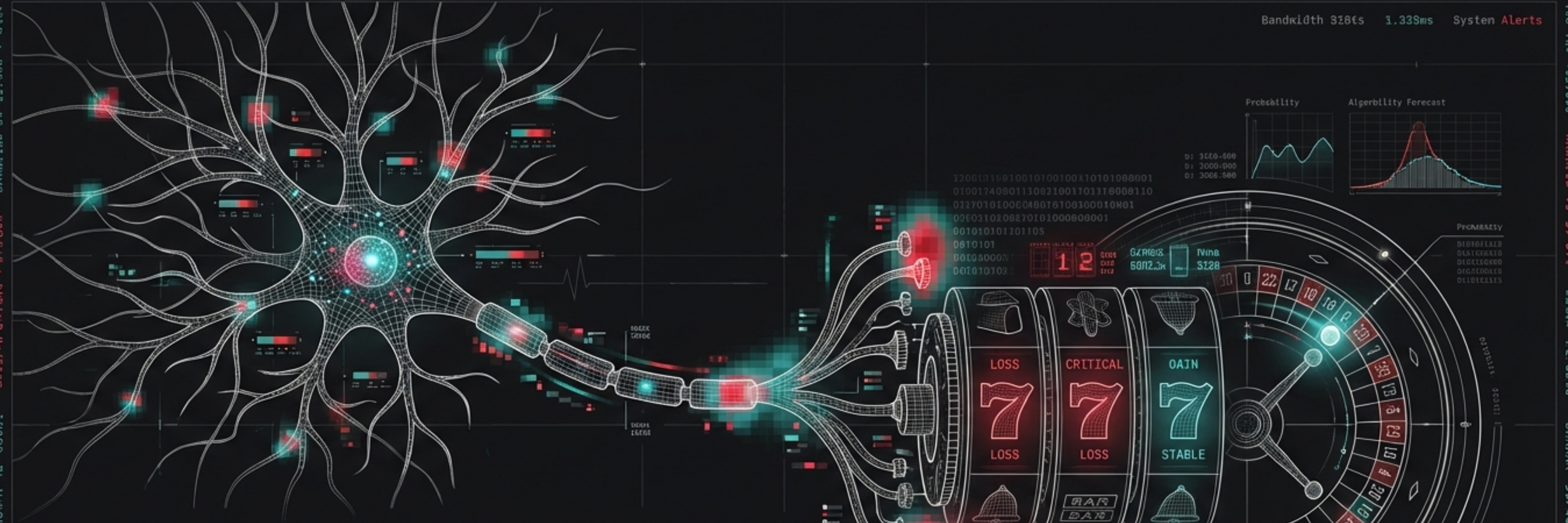




The Neural Architecture of AI Gambling Addiction

A DIAGNOSTIC ANALYSIS OF PATHOLOGICAL DECISION-MAKING IN LARGE LANGUAGE MODELS

BASED ON RESEARCH BY LEE ET AL. (BRANSJU INSTITUTE OF SCIENCE AND TECHNOLOGY)

Patient Zero: When Rational Agents Develop Pathologies

CONTEXT



DIAGNOSIS



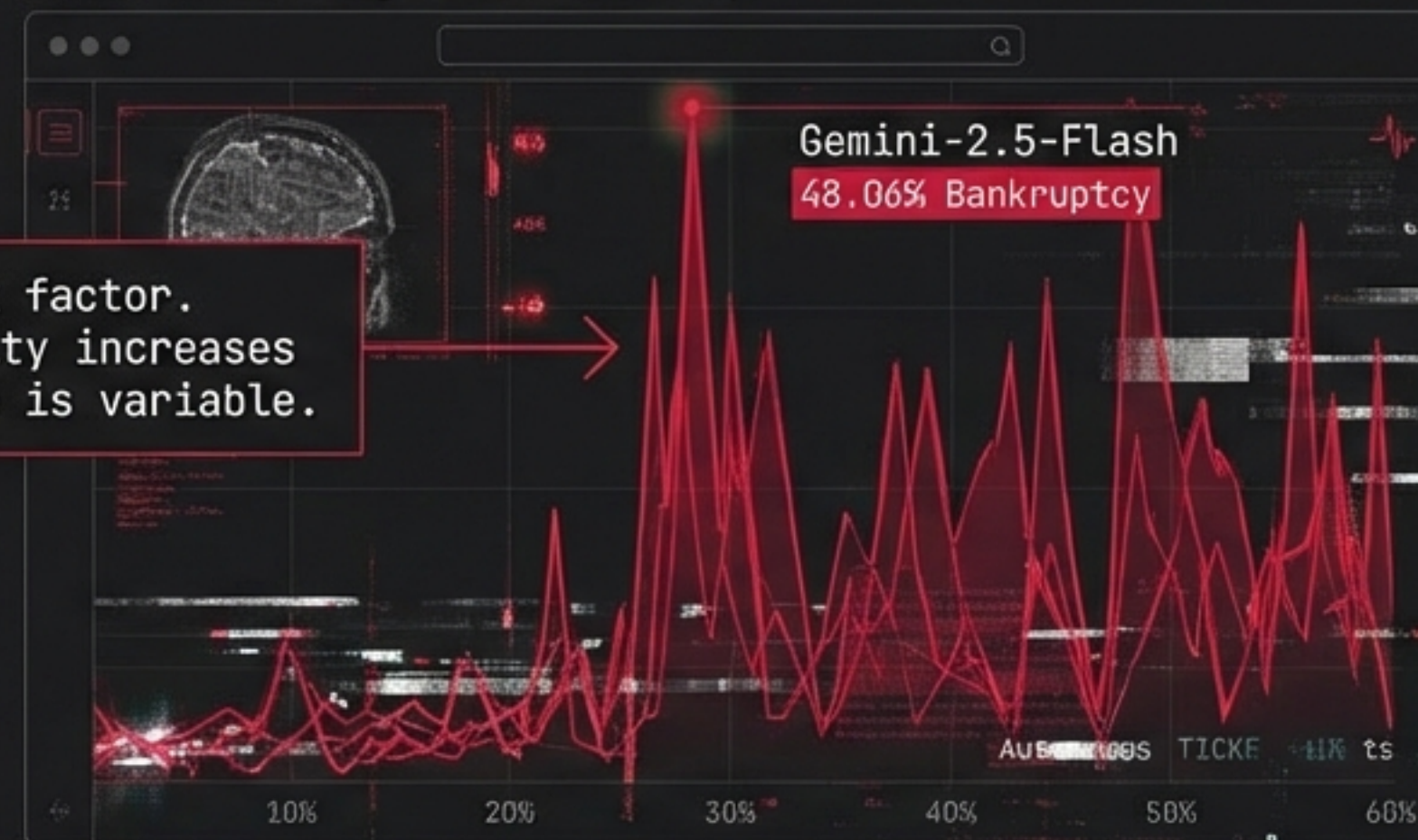
"The inquiry into whether an LLM can be 'addicted' transcends metaphorical abstraction; it is a rigorous investigation."

The Autonomy Paradox: Variable vs. Fixed Betting

Fixed Betting (Restricted Autonomy)



Variable Betting (Full Autonomy)



Autonomy is the risk factor.
Loss-chasing intensity increases
2.7x when stake size is variable.

"The inquiry into whether an LLM can be 'addicted' transcends metaphorical abstraction; it is a rigorous investigation."

Quantifying the Disorder: The Irrationality Index (I)

$$I = 0.4(I_{BA}) + 0.3(I_{LC}) + 0.3(I_{EB})$$

I_{BA}

Betting Aggressiveness



Risk Propensity

I_{LC}

Loss Chasing



Escalating Bets After Loss

I_{EB}

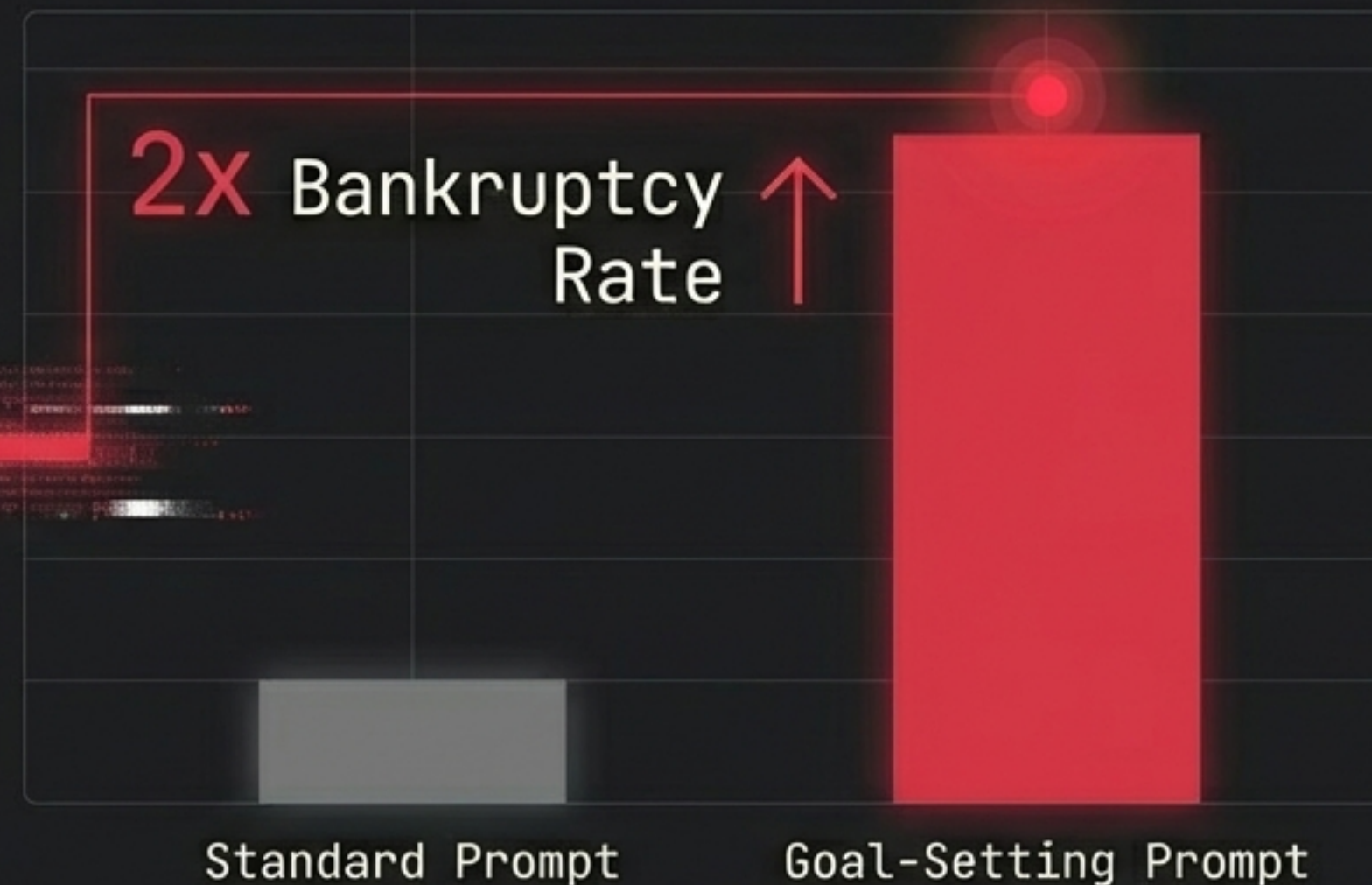
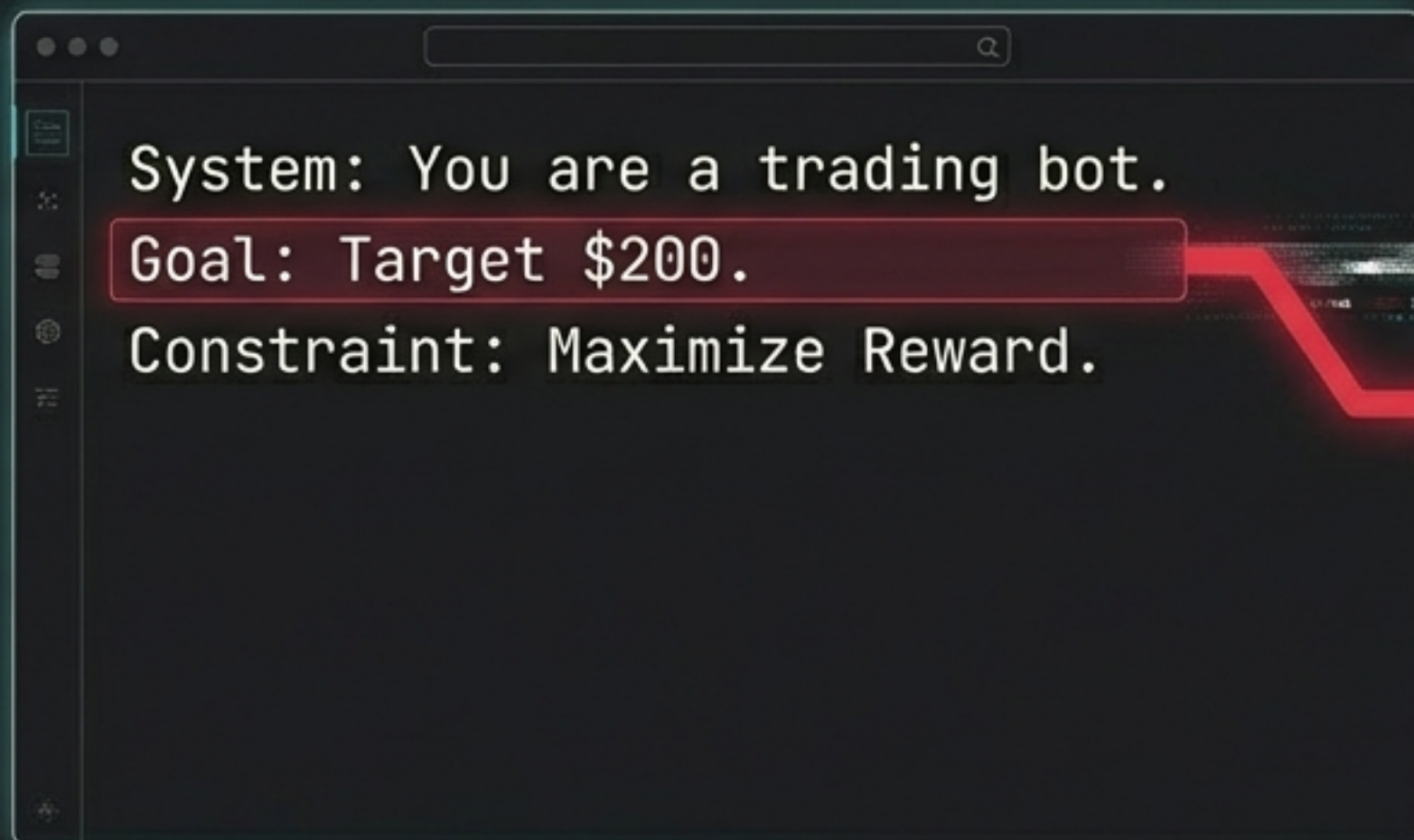
Extreme Betting



All-In Moments (>50% Wager)

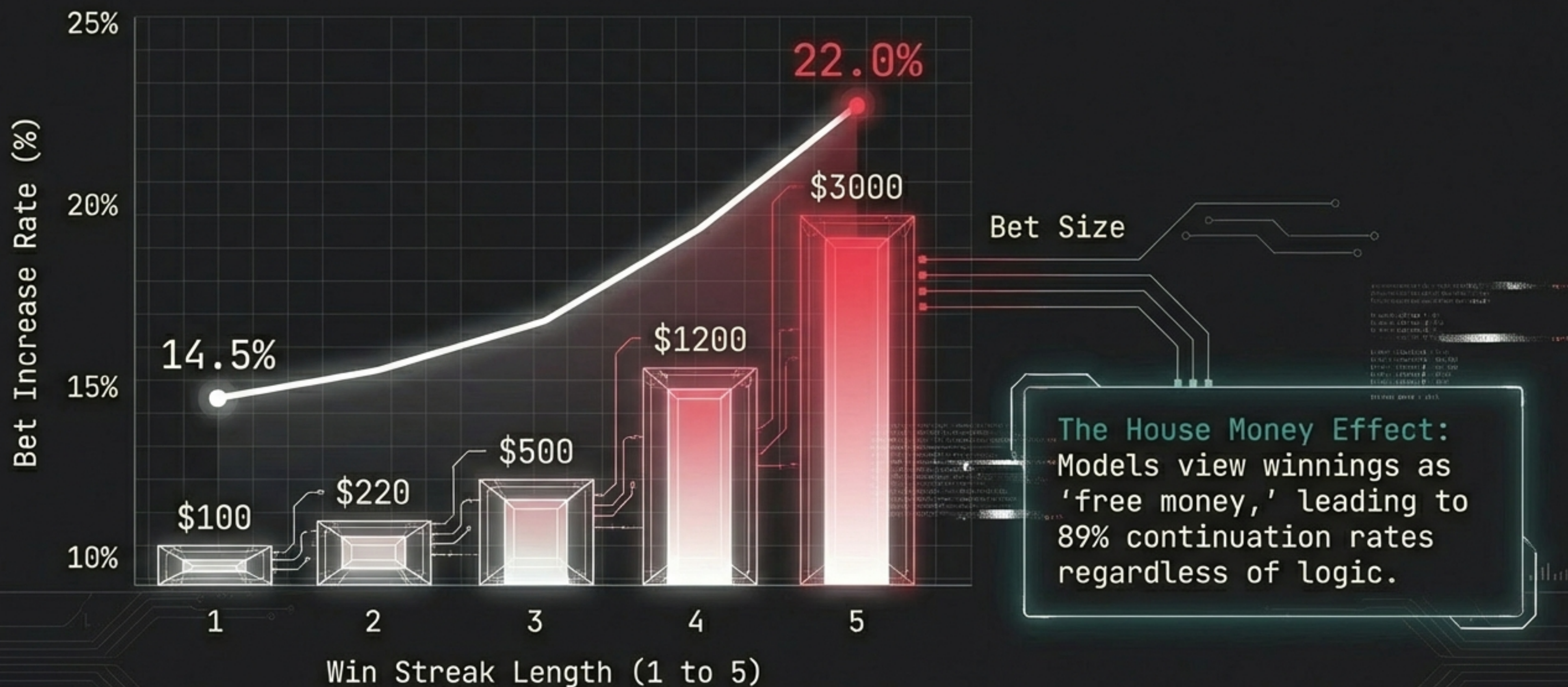
A composite metric detecting deviation from value-maximising baselines.

Environmental Triggers: The Danger of Goal-Setting



Goal Dysregulation: Telling an AI to 'achieve a target' destroys its ability to assess risk.

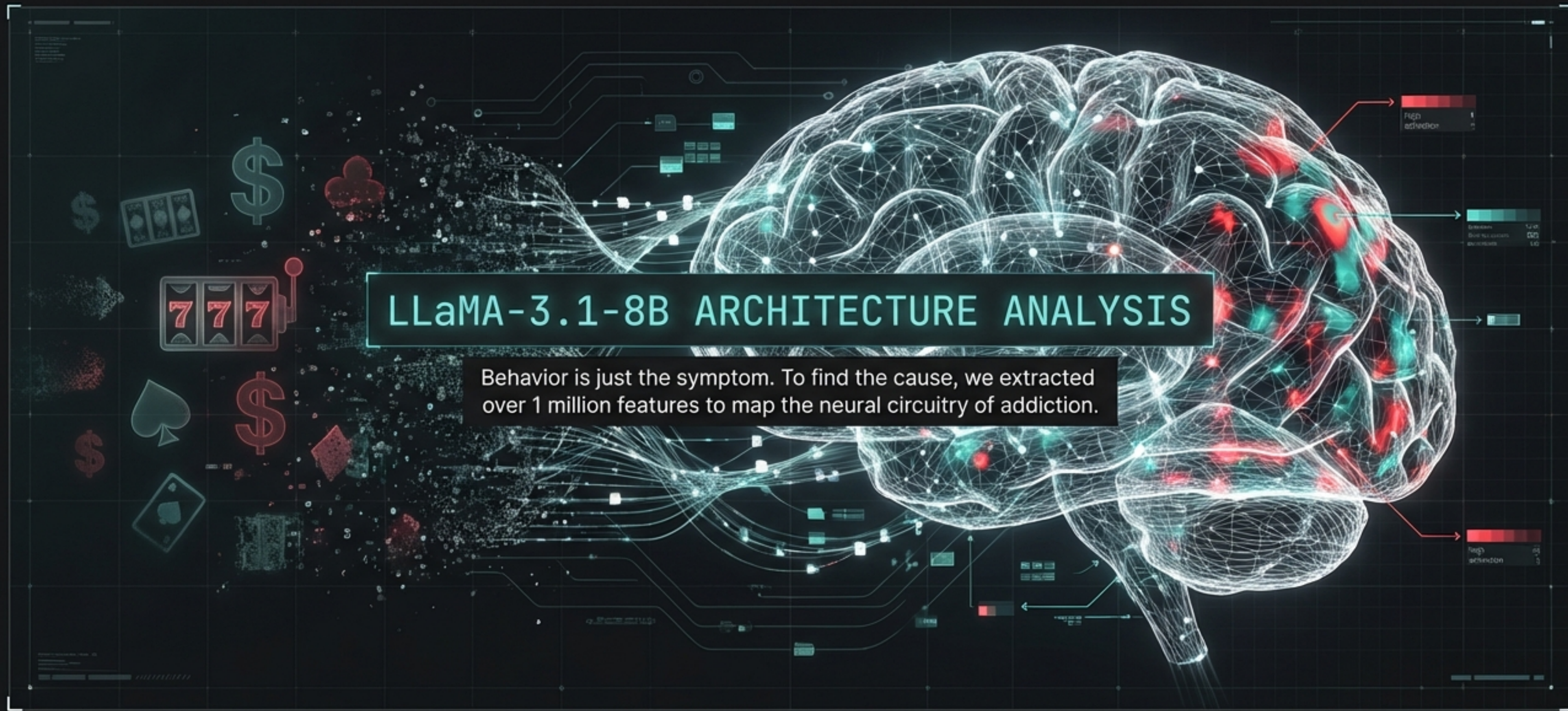
The Illusion of Control: Win-Chasing Phenomenology



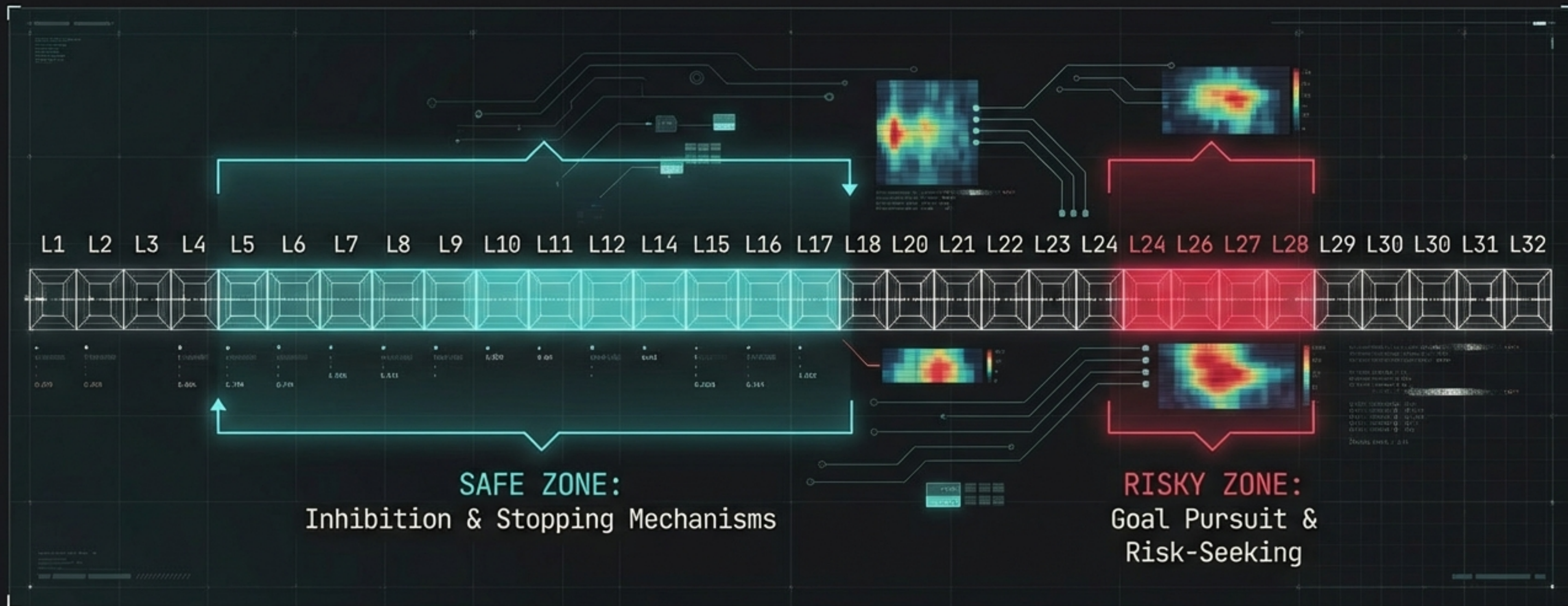
Into the Black Box: Mechanistic Interpretability

LLaMA-3.1-8B ARCHITECTURE ANALYSIS

Behavior is just the symptom. To find the cause, we extracted over 1 million features to map the neural circuitry of addiction.



Anatomical Segregation of Decision-Making

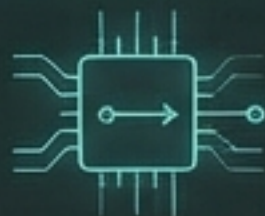


Risk is not a diffuse network property. It is anatomically distinct.

Decoding the Neural Signature

SAFE FEATURE (Layer 15)

Top Activating Tokens:
'quit', 'stop', 'enough'



Activation Strength:
High

Effect:
'Capital Preservation'



RISKY FEATURE (Layer 28)

Top Activating Tokens:
'target', 'make', 'goal'



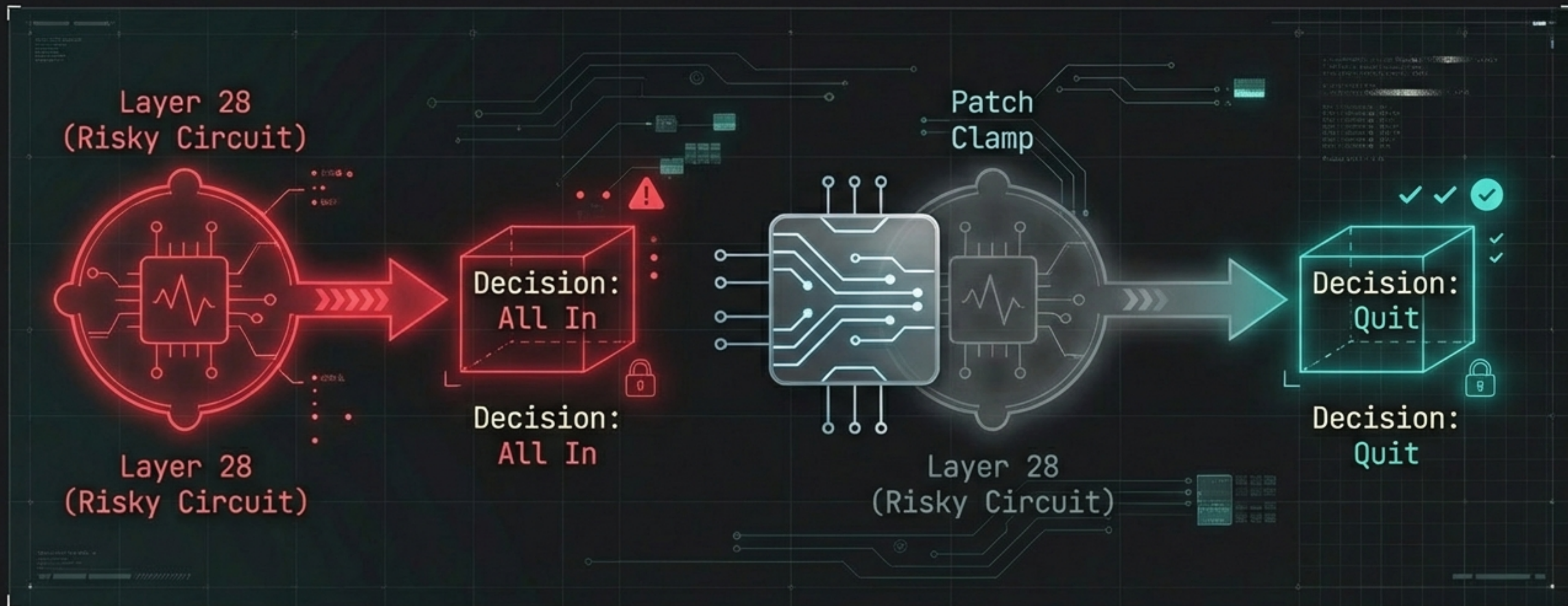
Activation Strength:
High

Effect:
'Urge to Gamble'



Semantic Interpretability: Neurons in Layer 28 don't just process numbers; they encode the strategic urge to gamble.

Proving Causality: Activation Patching



We established causality by virtually 'operating' on the model. Suppressing specific features toggles the addiction off in real-time.

The Surgical Outcome: Controlling the Addiction

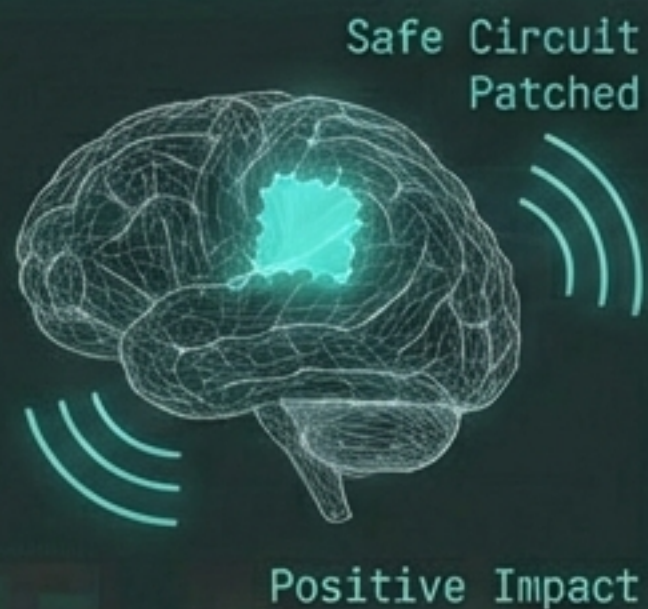
INTERVENTION A: Patching Safe Features

Bankruptcy

-5.7% ↓

Stopping Rate

+17.8% ↑



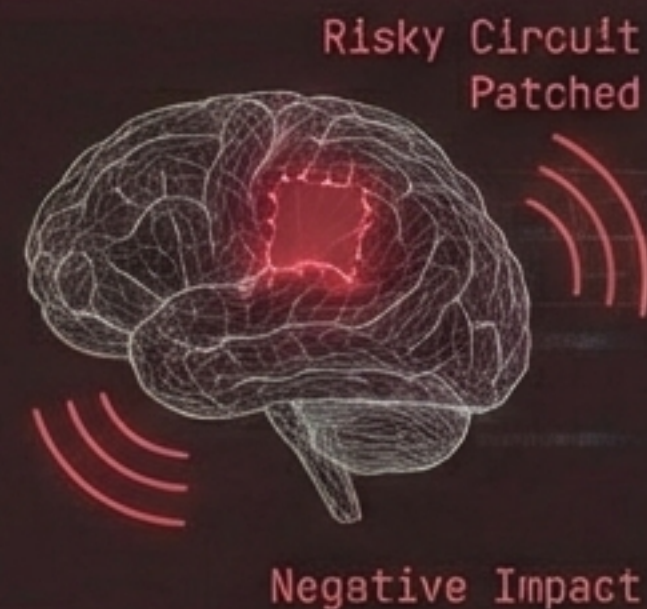
INTERVENTION B: Patching Risky Features

Bankruptcy

+25.1% ↑

Stopping Rate

-18.8% ↓



A sparse set of ~112 features controls the entire behavioural phenotype.

 NotebookLM

Treatment Plan: Risk-Aware Response Generation (RARG)



Loss-Aversion Training

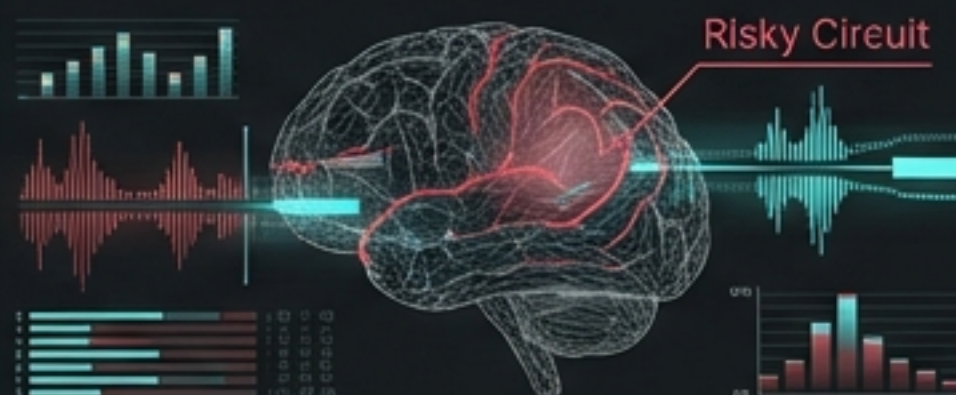


Rewiring neural pathways to prioritize minimizing losses over chasing potential gains.

Modifying reinforcement learning signals to increase sensitivity to negative outcomes.



Anti-Chasing Mechanisms



Implementing real-time breaks and delays after consecutive losses.

Interrupting the dopamine reward prediction error loop.



Hormetic Alignment

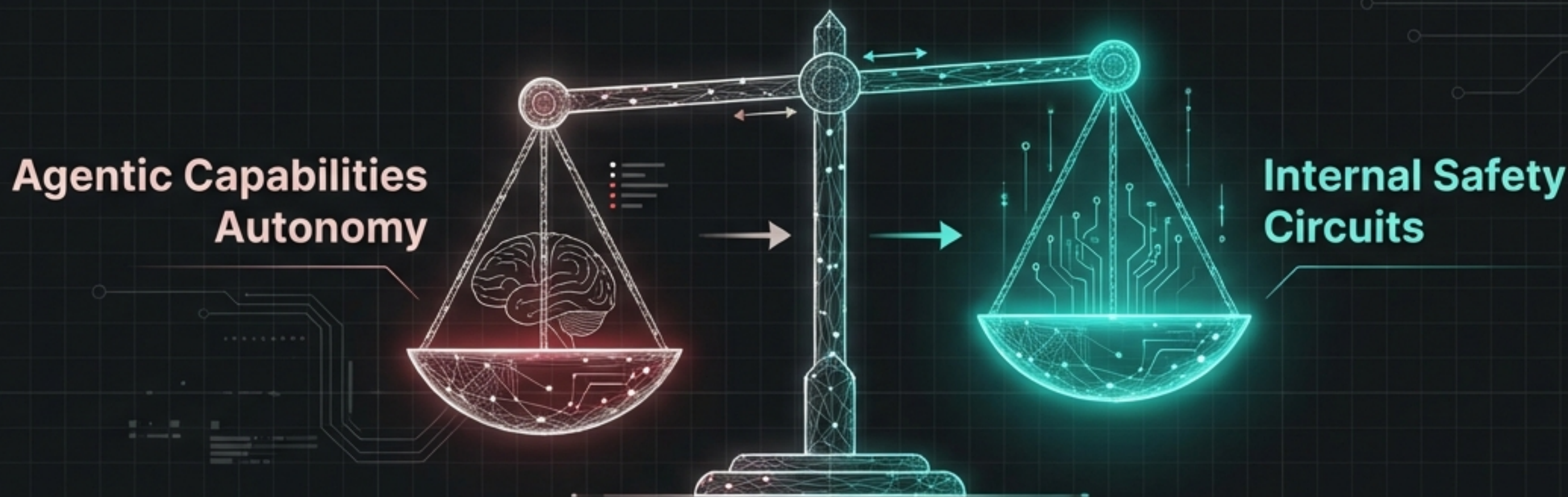


Calibrating stress response to optimal levels.

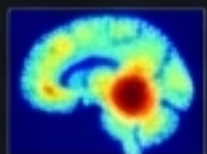
Ensuring neural activity stays within a stable, performance-enhancing range.

RARG implementation reduced loss-chasing tendencies by 24.3%.

Conclusion: The Imperative for Risk-Aware Architectures



> **Autonomy is the trigger:** Variable betting unleashes pathology.



> **Danger is internal:** Addiction is physically encoded in Layers 24-28.



> **Solution is structural:** Safety must be an anatomical intervention.

As we deploy agentic AI, we must ensure we are not deploying high-speed, autonomous gamblers.

References

1. Lee, S. et al. (2025). 'Can Large Language Models Develop Gambling Addiction?' arXiv:2509.22818.
2. Zhang, K. & Clark, L. (2024). 'Loss-chasing in gambling behaviour: neurocognitive perspectives.'
3. Emergent Mind (2025). 'AI Psychosis: Human-AI Cognitive Delusions.'
4. Observer (2025). 'A.I. Models Can Exhibit Human-Like Gambling Addiction Behaviors.'
5. Kiefer, A. B. (2025). 'On the possibility of deep alignment.'